

CLASSIFIED — DESIGN DOCUMENT

# BIOFALL

## CONTAINMENT FAILURE PROTOCOL

A top-down containment shooter. You are not a savior — you are a disposable operator pushed into a sealed city already lost to a bio-organic outbreak. Every mission is a controlled descent into systems already failing.

|                                              |                                              |
|----------------------------------------------|----------------------------------------------|
| <div>GENRE</div> <div>TOP-DOWN SHOOTER</div> | <div>ENGINE</div> <div>UNREAL ENGINE 5</div> |
| <div>SCRIPTING</div> <div>BLUEPRINT</div>    | <div>PLAYERS</div> <div>SOLO (MVP)</div>     |
| <div>PLATFORM</div> <div>PC</div>            | <div>STATUS</div> <div>PROTOTYPE</div>       |

01

## HIGH CONCEPT

A top-down containment shooter set inside a quarantined city overwhelmed by bio-organic infection. The focus is not survival, but controlled penetration into unstable zones where systems collapse in real time.

The game should feel overwhelming but always legible. Clean architecture is consumed by organic growth; the player pushes forward through collapsing sectors toward extraction.

02

## CORE PILLARS

P 1

### PRESSURE OVER COMPLEXITY

Tension comes from density and pacing, not from system depth.

P 2

### READABLE CHAOS

The screen can be overwhelming but must always be legible.

P 3

### WEAPON FEEL FIRST

Gunplay feedback is the foundation everything else builds on.

P 4

### SHORT, REPEATABLE MISSIONS

Clear escalation, fast loop, finish over expand.

03

## PITCH & FANTASY

### ELEVATOR PITCH

A response unit is sent into a dead city to stabilize what cannot be saved. Every mission is a push into systems already failing.

### PLAYER FANTASY

You are not a hero. You are a system response — efficient, replaceable, and constantly under pressure. Efficiency matters more than survival.

04

## tone, style & world

Serious, grounded, oppressive. No humor, no parody. Visual tone relies on one contrast: **sterile vs infected**.

A classified bio-reactive material escaped containment. It consumes structure and rewrites matter into functional biomass. The city is divided into sectors, each degrading differently depending on exposure level.

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## GAMEPLAY LOOP

01

DEPLOY

02

ADVANCE

03

STABILIZE

04

ESCALATION

05

EXTRACT

06

UPGRADE

Death is not a fail-state with consequence layering — **death restarts the mission**. The loop stays tight and repeatable.

06    **CORE MECHANICS**

|             |                                                       |
|-------------|-------------------------------------------------------|
| MOVEMENT    | Top-down traversal with constant forward pressure.    |
| COMBAT      | Aim + fire; weapons rely on recoil and spread.        |
| MOBILITY    | Limited dodge / sprint for repositioning only.        |
| INPUT       | Keyboard+Mouse and Gamepad treated as equal priority. |
| NO OVERLOAD | No ability stacking in MVP — clarity over options.    |

07    **ENEMY ROSTER**

Enemies are simple individually but dangerous in density.

|                                                                                               |                                                                                       |                                                                                    |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>WALKER</b><br><br>PRESSURE<br><br>Baseline threat. Applies constant, predictable pressure. | <b>RUNNER</b><br><br>DISRUPTOR<br><br>Breaks player positioning and forces reaction.  | <b>BRUTE</b><br><br>ABSORBER<br><br>Soaks damage and blocks paths. Controls space. |
| <b>SPITTER</b><br><br>ZONER<br><br>Ranged threat that forces movement and denies ground.      | <b>BLOATER</b><br><br>PUNISHER<br><br>Punishes proximity. Rewards spacing discipline. |                                                                                    |

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OBJECTIVES & EVENTS

| OBJECTIVES |                                 | DYNAMIC EVENT SYSTEM                                                                                                        |
|------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| SEAL       | Close contamination zones.      | Disruptions shift pacing without changing core rules: blackout zones, sudden surges, lockdown gates, environmental hazards. |
| ACTIVATE   | Restore infrastructure systems. |                                                                                                                             |
| SURVIVE    | Hold against escalating waves.  |                                                                                                                             |
| EXTRACT    | Reach the extraction point.     |                                                                                                                             |
| OPTIONAL   | Higher risk for higher reward.  |                                                                                                                             |

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LEVEL STRUCTURE & CAMERA

Each mission is a contained sector with 2–4 objectives. Layouts favor flow over exploration — the player is constantly pushed forward.

|             |                                                   |
|-------------|---------------------------------------------------|
| CAMERA      | Fixed angled top-down. High-contrast silhouettes. |
| LEVELS      | Handcrafted modular layouts. Density over size.   |
| READABILITY | Minimal clutter; effects readable under stress.   |

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ART DIRECTION

Rigid geometry overtaken by organic mass. Pulsating surfaces, slow environmental decay.

**Palette:** cold greys versus infected reds and greens. Sterile structure as the canvas; infection as the only color that draws the eye.

## EXAMPLE MISSIONS

**SECTOR ENTRY** Tutorial pressure — teach the loop under light load.

**POWER GRID** Defend and restore infrastructure under waves.

**CONTAINMENT LAB** Origin hint — narrative through environment.

**EVAC CORRIDOR** Sustained movement under pressure to extraction.

## PROGRESSION & ECONOMY

### PROGRESSION

Linear weapon unlocks. Practical upgrades only — damage, fire rate, mobility. No deep RPG layering.

### ECONOMY

Bio Samples collected per mission, spent on weapons and upgrades. The economy stays simple and functional.

MVP

|                                   |                                      |
|-----------------------------------|--------------------------------------|
| <div>1</div> <div>LEVEL</div>     | <div>1</div> <div>PLAYER</div>       |
| <div>3-4</div> <div>WEAPONS</div> | <div>4</div> <div>ENEMY TYPES</div>  |
| <div>1</div> <div>MINI-BOSS</div> | <div>1</div> <div>EXTRACT LOOP</div> |

Vertical slice target: one polished mission with full loop, enemy variety, and the complete feel of combat and pacing.

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TECHNICAL STACK

|        |                                                       |
|--------|-------------------------------------------------------|
| ENGINE | Unreal Engine 5 — Top Down template, Blueprint-first. |
| CAMERA | Engine camera rig, fixed angled top-down.             |
| DATA   | Data Assets / Data Tables for weapons & upgrades.     |
| PERF   | Actor pooling for enemies, projectiles, effects.      |
| VFX    | Niagara for infection, impacts, environmental decay.  |
| TARGET | PC. No early multi-platform complexity.               |

REVISION NOTE — ENGINE CHANGE

Project migrated from Unity 6 URP (original GDD) to **Unreal Engine 5 (Blueprint)**. All prior Unity-specific references (ScriptableObjects, Cinemachine, URP) are superseded by their Unreal equivalents above.

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CORE SYSTEMS & ARCHITECTURE

SYSTEMS

|           |                                  |
|-----------|----------------------------------|
| PLAYER    | Movement, aim, fire controller.  |
| WEAPONS   | Data-driven, pooled projectiles. |
| ENEMY AI  | State-based behavior.            |
| DIRECTOR  | Wave / spawn manager.            |
| OBJECTIVE | Mission + extraction flow.       |
| UI        | Minimal HUD; save/load.          |

ARCHITECTURE PRINCIPLES — NON-NEGOTIABLE

|           |                               |
|-----------|-------------------------------|
| OOP       | Polymorphism & Encapsulation. |
| SINGLETON | Shared managers.              |
| OBSERVER  | Event-driven comms.           |
| SOLID     | Applied throughout.           |
| POOLING   | Enemies, projectiles, FX.     |

ENEMY STATE MACHINE — CLARITY OVER COMPLEXITY



**HUD:** health, ammo, objective. No clutter. Information must be readable instantly under pressure.

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VERSION CONTROL & PIPELINE

Solo project. Only one person ever edits assets, even when in-game co-op (2–4 players) is added much later.

|          |                                                           |
|----------|-----------------------------------------------------------|
| VCS      | Git + GitHub + Git LFS for binary assets.                 |
| PERFORCE | Not required — no concurrent binary asset editing.        |
| CO-OP    | A gameplay feature (multiplayer), not team collaboration. |
| TOOLS    | Blender (3D), Figma & GIMP (2D/UI).                       |

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## WEEK 01

**PROTOTYPE**

Movement, aim, fire. Core feel established.

## WEEK 02

**ENEMY + COMBAT**

AI states, enemy roster, weapon feedback.

## WEEK 03

**OBJECTIVES**

Mission flow, waves, extraction loop.

## WEEK 04

**POLISH + TEST**

Readability pass, feel tuning, playtest.

RISKS

|         |                                     |
|---------|-------------------------------------|
| GUNPLAY | Weak combat feel.                   |
| DENSITY | Low enemy pressure.                 |
| SCOPE   | Over-engineering systems.           |
| NOISE   | Visual clutter killing readability. |

FINAL DIRECTION

Keep systems minimal. Push density and feel. Cut features early. Finish a small but polished experience instead of expanding scope. Atmosphere comes from pressure — not narrative.

END OF DOCUMENT — CONTAINMENT HOLDS